
HPL *ink* IPTV Enigma2

User manual

A guide to the plugin for Enigma2 set-top boxes. The full English manual follows; the Polish manual is published separately.

Platform

Enigma2 — OpenBH 5.x (Python 3.12) and OpenBH 6.0 (Python 3.14)

Release

v2.01 04.10.2026

Sources

M3U / M3U8 lists, Xstream Codes, XMLTV EPG

Control

box remote + browser panel

Labels: **TV** means an action on the TV with the box remote. **PANEL** means an action in a browser on a computer or phone, at the address shown on the TV.

Document release: this manual describes **HPL ink IPTV v2.01** of 04.10.2026. The plugin is **free** and needs no activation. The plugin and panel language (Polish or English) is chosen in Settings.

English user manual

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1. What the plugin is

HPL ink IPTV is a plugin for satellite set-top boxes running Enigma2. It plays channels, movies and series from IPTV lists, shows the TV guide and reminds you about programmes. Lists, the guide and settings are easiest to add from a computer or phone, in the browser panel.

- M3U / M3U8 lists from a URL, from a file on your computer or from an Xtream Codes account (server, login, password).
- TV channels in three columns: bouquets, channels and a card with the current programme.
- Its own player with OSD, channel list, tile strip, reminders and automatic recovery of a stopped picture.
- Movies with a poster wall, series with the next episode, resuming from where you stopped.
- XMLTV guide with automatic import, guide search, reminders with a choice of „Remind“ or „Switch“.
- Favourites, history, a „Watch later“ list, Polish and English language.

Network: the box, computer and phone must be in the same home network. The panel is not a website on the internet — it works only at home, at the address shown on the TV.

2. Hardware and software

What the plugin was tested on

Set-top box	Software	Python
VU+ Duo 4K SE	OpenBH 5.x	Python 3.12
VU+ Solo 4K	OpenBH 5.x	Python 3.12

In addition, the plugin was checked on **OpenBH 6.0 (Python 3.14)** — it works correctly after installing one package, `python3-sqlite3` (chapter 3).

Playback was checked on both Enigma players: **GStreamer (4097)** and **exteplayer3 (5002)** with the ServiceApp plugin.

Other boxes and images: the plugin is written in pure Python and uses standard Enigma2 components. In theory it should also work on other boxes and other Enigma2-based images with **the same Python version (Python 3.12 or 3.14)**, e.g. other VU+ models or other images of the same generation. Such combinations have not been tested, though.

Images with the older Python 2.7 (e.g. OpenATV 6.x) are not supported by this version of the plugin.

Requirements

- An Enigma2 box connected to the home network (cable or Wi-Fi).
- An IPTV list from your provider: an M3U address, an M3U file or Xstream Codes account details.
- Access to the box over FTP / SFTP (e.g. WinSCP, FileZilla) and Telnet / SSH (e.g. PuTTY) — only for installation.
- The `python3-sqlite3` Python module — included in the OpenBH 5.x image, **on OpenBH 6.0 it must be installed** (chapter 3).

3. Installation

1. File

Copy `enigma2-plugin-extensions-hpliptv_2.01_all.ipk` to the `/tmp` folder on the box (FTP / SFTP).

2. Install

Log in to the box over Telnet / SSH and enter the install command (below).

3. Restart GUI

Restart the interface:
Menu → Standby / Restart → Restart GUI.

4. Start

The plugin is in *Menu → Plugins* and in the extensions menu (Extensions) as **HPL ink IPTV**.

Install command:

```
opkg install /tmp/enigma2-plugin-extensions-hpliptv_2.01_all.ipk
```

Updating to a newer version works the same way — the new package replaces the old one, settings and lists stay.

Uninstall: `opkg remove enigma2-plugin-extensions-hpliptv`.

OpenBH 6.0 — an additional package is required. The OpenBH 6.0 image does not include the Python database module in which the plugin stores the EPG guide. Without it the EPG import ends with the message „*The sqlite3 module is missing - EPG has nowhere to be saved*“. After installing the plugin, before the first EPG import, enter over Telnet / SSH:

```
opkg update && opkg install python3-sqlite3
```

and restart the GUI. In OpenBH 5.x this package is already in the image — nothing needs to be installed. To check whether the package is present: `opkg list-installed | grep python3-sqlite3`.

exteplayer3 player (optional)

By default the plugin plays through **GStreamer (4097)**, which every image has. The second player, **exteplayer3 (5002)**, is sometimes better with high-bitrate channels. It needs the ServiceApp plugin from your image's repository:

```
opkg update && opkg install enigma2-plugin-systemplugins-serviceapp
```

After restarting the GUI choose *Player: exteplayer3 (5002)* in the plugin Settings. When ServiceApp is not installed, the plugin plays through GStreamer on its own and says so in a message.

ServiceApp must match the system. Install it only from the repository of the same image you have on the box. A package from another image or an older archive may fail to load and give a black screen on exteplayer3.

4. First start

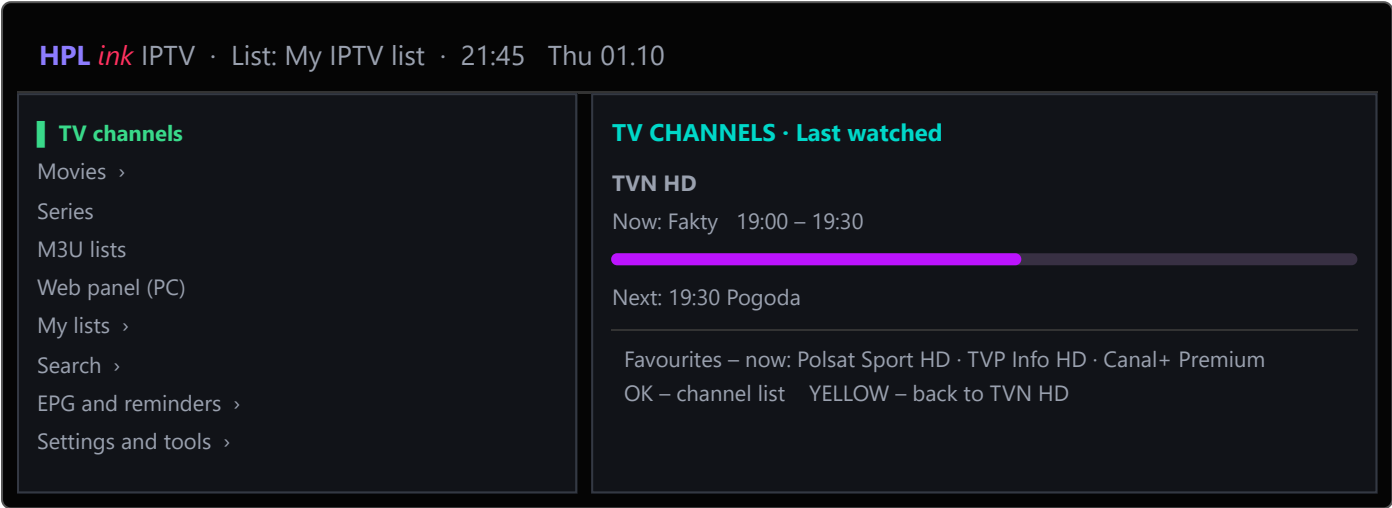
On the first start the plugin has no list yet. The simplest way is to add one through the browser panel.

1. In the plugin choose **Web panel (PC)** **TV**. The screen shows the panel address, e.g. `http://192.168.1.50:8098`, a QR code and an access key.
2. Open this address in a computer browser or scan the QR code with a phone **PANEL**.
3. In the panel go to **Settings** → **Add list** (or the **Add list** button on the Home page) and enter the list address or the Xstream Codes details. A file from your computer is added in the same place: **Upload an .m3u file from the computer**.
4. The list downloads to the box. Go back to the plugin on the TV — channels, movies and series are ready.
5. The EPG guide downloads on its own („Auto PL” source). It can also be imported right away: **EPG and reminders** → **Import TV EPG**.

Without a computer: the list address can also be entered on the TV, in **Settings and tools** → **Settings** → **Source 1**, with the **YELLOW** key (edit), then choose **Refresh M3U list**.

5. Start screen

On the left there is a list of nine items, on the right a preview panel with the data of the selected item. At the top: clock, date and the name of the active list; at the bottom: the number of channels, movies and series.



Item	What the preview panel shows	YELLOW key
TV channels	the last watched channel with the current and next programme, favourite channels with what is on now	plays the last channel
Movies >	a stopped movie with a progress bar, the „Watch later“ list	resumes the movie where it stopped
Series	series in progress with the next episode	plays the next episode
M3U lists	the active list, when it was downloaded, how many items it has	—
Web panel (PC)	panel address and access key	—
My lists >	the last watched item and counters	—
Search >	search types	—
EPG and reminders >	guide range, last import, upcoming reminders	—
Settings and tools >	version and key settings	—

Items marked > open a submenu.

OK

 enters an item,

EXIT

 closes the plugin. After you stop watching you return to the same submenu you came from.

6. TV channels

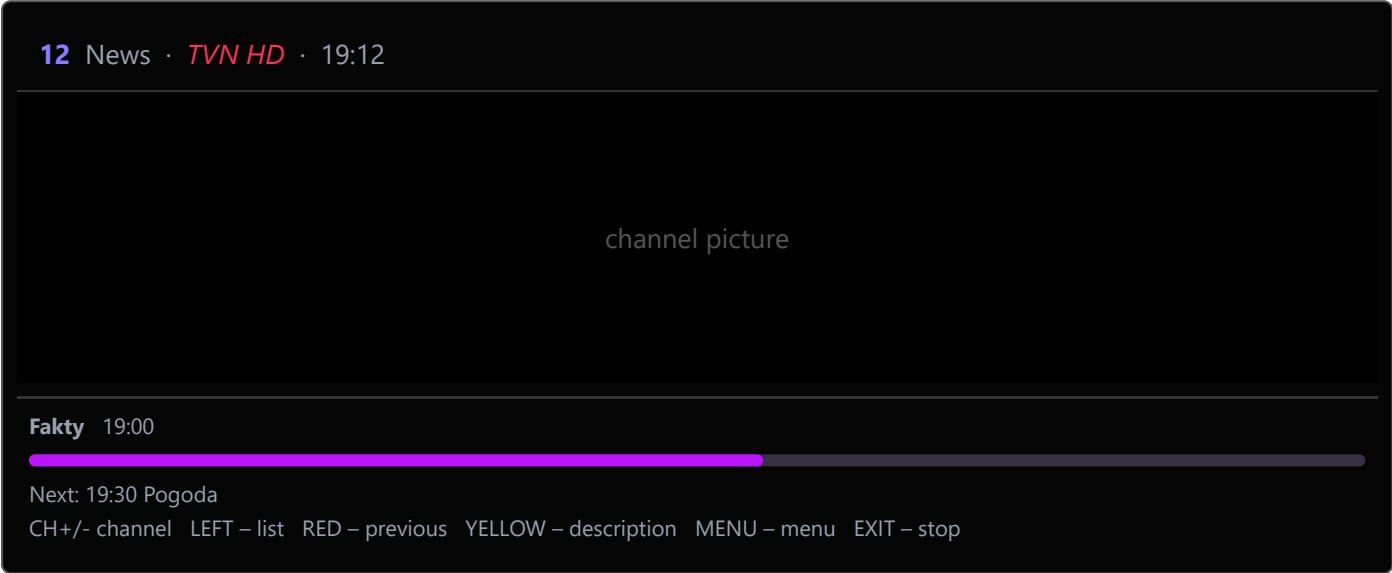
The screen has three columns: **bouquets** (groups from the list, „Search channel“ at the top), **channels** of the selected bouquet and a **card** of the selected channel — logo, current programme with times and a progress bar, description and the next programme. The channel that is playing has a play marker.

Key	Action
LEFT / RIGHT	move between columns
UP / DOWN , CH+ / CH-	move in the list, page up / down
OK	opens a bouquet or plays a channel; on the button under the card — a reminder about the next programme
MENU	channel menu: Channel guide, Add to / Remove from favourites, Remind about the next programme, Switch to the next programme, Programme description
INFO	full description of the current programme
YELLOW	adds the channel to favourites / removes it
BLUE	search a channel by name
GREEN	EPG guide import
EXIT	back

The **channel guide** shows the schedule for the next days. **OK** opens the programme description, **GREEN** sets a reminder, **YELLOW** sets a channel switch when the programme starts. The programme gets a *REMIND.* or *SWITCHES* marker. The same key again deletes the reminder, the other colour changes its type.

7. Player

When you play a channel or a movie, the OSD appears: channel number and group, name, clock, logo, programme title, start and end time, a purple progress bar, the next programme and key hints. The OSD hides by itself after the time set in *OSD bar time*.



Key	Action
CH+ / CH-	next / previous channel in the bouquet
LEFT	channel list of the bouquet over the picture: OK switches, CH+/- page, LEFT / EXIT closes
RIGHT	tile strip with the next channels (logo, name, what is on now): LEFT / RIGHT move, UP programme description, OK switches
RED	back to the previous channel
YELLOW	full programme description (UP / DOWN scrolls)
MENU	player menu (on the right side of the screen)
AUDIO / TEXT	audio track / subtitle selection
EXIT	hides the OSD when it is visible, stops playback when it is hidden
playback keys	with movies: pause and seeking as in the Enigma player

Player menu — MENU key

- **Programme description** and **Channel guide** — without stopping playback, the picture keeps playing underneath.
- **Remind: HH:MM title** or **Switch: HH:MM title** — a reminder about the next programme on this channel; when one is already set — **Delete reminder**.
- **Reminders** — the list of all scheduled reminders, on a semi-transparent background.
- **History** — recently watched items, switching without leaving the player.

Notices and reminders on the picture

- **Small progress bar** in the bottom-right corner when the OSD is hidden (can be switched off).
- **Next programme announcement** a chosen number of minutes before it starts.
- **„Later” panel** — three upcoming programmes near the end of the current one.
- **Reminder** — a bar in the bottom-right corner: „In 10 min · channel”, and at the start „Starting · channel” or „Switching · channel”. When the plugin is not playing, the reminder appears in an Enigma window. A waiting reminder is shown by a dot in the OSD.

When the picture stops

The plugin watches the stream by itself. When the picture stands still for more than a dozen seconds, the message „Restoring picture...” appears and the plugin tries to rebuild the connection — up to three times. If that does not help and the list has a second channel with the same name (a backup link), the plugin switches to it. When the network is down, it waits for it to return.

Movies and series in the player

A movie stopped halfway remembers its position — when you play it again, the plugin offers to continue. After a series episode ends, the next one starts.

8. Movies and series

Movies

The **Movies** submenu has two forms: **Poster wall** (a grid of covers, three sizes in Settings) and **List** (a plain list of titles).

Key	Poster wall	Movie card
OK	movie details card	play
GREEN	—	audio track selection
YELLOW	add to / remove from favourites	subtitle selection
BLUE	filter by name	add to / remove from „Watch later“
CH+ / CH-	page of posters	—
EXIT	back	back

The movie card shows the poster, category, rating and description (when available in the list or in the TMDb database).

Series

Series are grouped from the episodes in the list. Choose a series, then an episode. The plugin remembers watched episodes and suggests the next one — also on the start screen (**YELLOW** key).

9. My lists and Search

My lists

- **Continue watching** — movies and episodes stopped halfway.
- **Favourites** — channels and items marked with the **YELLOW** key.
- **Watch later** — movies put aside for later.
- **Recently watched** — history.

Search

- **Search movie/entry** — in movies and series from the list, with the on-screen keyboard.

- **Search the programme guide (EPG)** — in programme titles of the whole guide, from now on. In the results:
OK plays the channel of a running programme, GREEN sets a reminder, YELLOW a channel switch.

Channels are searched on the channel list, with the BLUE key.

10. EPG and reminders

EPG guide

The plugin uses a guide in XMLTV format. The source is chosen in Settings:

- **Auto PL** (default) — a set of Polish guides from the internet.
- **url-tvg from the M3U list** — the guide address stored in the list itself.
- **Own XMLTV URL** (EPG custom URL) — any address of an `.xml` or `.xml.gz` file.

With an Xstream Codes list the guide is set automatically from the Xstream server when the list gives no address of its own. The import runs automatically every 48 hours and does not disturb watching — while something is playing, it waits. Manually: **EPG and reminders** → **Import TV EPG**.

EPG matching

When a channel has no schedule while others do, assign the right guide channel to it: **EPG and reminders** → **EPG matching**. `OK` assigns, `BLUE` filters by name, `YELLOW` deletes the assignment. The same can be done in the panel, in **Settings** → **EPG guide** → **EPG matching**.

Reminders

A reminder is set on a future programme: in the channel guide, in the guide search results, in the channel menu, in the player menu or in the panel. Every reminder has a type:

Type	What happens when the programme starts	Where to set it
Remind	only a message on the picture	<code>GREEN</code> <code>TV</code> , „Remind” button <code>PANEL</code>
Switch	the plugin switches to the programme's channel	<code>YELLOW</code> <code>TV</code> , „Switch” button <code>PANEL</code>

The list of all reminders: **EPG and reminders** → **Reminders** or `MENU` in the player. `RED` or `YELLOW` deletes the selected one. Reminders that will switch the channel have a *SWITCHES* marker. An earlier advance notice (5–30 minutes) is set in Settings.

11. Browser panel

The panel is a page served by the box itself. The address and access key are shown by the **Web panel (PC)** item on the start screen — with a QR code for a phone. The address looks like `http://box-IP:8098`. The address with the key should not leave your home.

Tabs

Tab	Contents
Home	„Search everywhere“ (channels, movies, series, programmes), Back to it, Reminders, Continue watching, Watch later, Your series, Favourite channels, Tonight, Recently added, Recently watched; the M3U lists card (choosing the active list) and Automatic download
Channels	bouquets and channels with the current programme; a click plays the channel on the box; channel schedule window
Guide	schedule grid (earlier / now / later), guide search, scheduled reminders
Movies / Series	catalogue with posters, details, playing on the box, episodes
History	favourites, history, watch later, resume points
Settings	sub-tabs: Add list (address, Xstream, file from the computer), Bouquets (bouquets and catalogues to hide), EPG guide (source, import, EPG matching), Plugin (plugin settings)

TV bar

At the top of the panel there is a bar showing what plays on the box, with control buttons: previous / next channel, seeking, pause, stop, volume and mute.

Reminders from the panel

In the channel schedule window (**Channels** tab → schedule) every future programme has two buttons: **Remind** and **Switch**. A running programme has **Play**. A reminder that is set can be deleted in the same place.

M3U lists

1. **Settings** → **Add list**.
2. Type: *M3U list address* (name + `http://...` address) or *Xstream Codes* (server `http://server:port`, login, password).
3. **Add**. The list downloads to the box. The active list is chosen on the **Home** page, in the **M3U lists** card.

A file from your computer: **Upload an .m3u file from the computer**. Automatic download of the active list once a day is switched on on the Home page, in the **Automatic download** card. The active list can also be changed on the TV: **M3U lists** on the start screen.

12. Remote keys — summary

Screen	OK	Colours	Other
Start screen	enter	YELLOW quick return	EXIT closes the plugin
TV channels	bouquet / play	GREEN EPG, YELLOW favourites, BLUE search	MENU channel menu, INFO description, LEFT/RIGHT columns
Player	—	RED previous, YELLOW description	CH± , LEFT list, RIGHT tiles, MENU , AUDIO , TEXT , EXIT
Channel guide	description	GREEN remind, YELLOW switch	EXIT back
Guide search	play channel	GREEN remind, YELLOW switch	EXIT close
Poster wall	details	YELLOW favourites, BLUE filter	CH± page
Movie card	play	GREEN audio, YELLOW subtitles, BLUE watch later	EXIT back
Reminders	close	RED / YELLOW delete	EXIT exit
Settings	edit field	GREEN save, YELLOW edit text, BLUE clear cache, RED clear guide	LEFT/RIGHT change value, EXIT cancel

13. Settings

Settings and tools → **Settings** **TV** or the **Settings** tab **PANEL**. On the TV the settings are in five sections. A value is changed with **LEFT** / **RIGHT** and saved with **GREEN**.

Playlists and sources

1. **M3U sources file / Active source from file** — the lists added in the panel and the choice of the active one.
2. **Source 1–5 (file or URL)** — backup profiles with a list address, used when no list was added in the panel; **Active backup profile** picks one of them.
3. **M3U list - auto download** — „Once a day” or „No - on demand only”.

Playback

1. **Player** — GStreamer (4097) or exteplayer3 (5002, needs ServiceApp).
2. **Preferred audio language** — Polish, English, German or unchanged.
3. **Subtitles at start** — off or unchanged.

EPG and reminders

1. **EPG source** and **EPG custom URL (XMLTV)** — see chapter 10.
2. **EPG range (days)** — 2, 3, 5 or 7 days of schedule.
3. **EPG auto-import** — import every 48 hours or only manually.
4. **Next programme announcement (min)** — how many minutes before the start to show the notice (or off).
5. **Send EPG to Enigma** — also passes the schedule to Enigma's own guide.
6. **Reminders** — the default type for reminders without a chosen type: „Message only” or „Switch channel”.
7. **Reminder advance notice** — 5, 10, 15 or 30 minutes earlier, or off.

Appearance and language

1. **Language** — Polish or English (plugin and panel).
2. **Channel logos** — first from the EPG guide or first from the M3U list.
3. **OSD bar time (s)** — 5, 10, 15 or 20 seconds.
4. **Small progress bar (no OSD)** and **„Later” panel (3 programmes)** — on / off.

5. Channel list font, Programme description font — three sizes.

6. Poster size in Movies — small, medium or large.

Web panel (PC)

1. Web panel (PC) — switches the browser panel on or off.

Tools

- **Refresh M3U list** — downloads the active list again.
- **Save list to .m3u** — saves the downloaded list locally on the box.
- **Diagnostics** — the state of the list and the guide, useful when reporting a problem.

14. Troubleshooting

Message „The sqlite3 module is missing - EPG has nowhere to be saved“

The `python3-sqlite3` package is missing — this concerns OpenBH 6.0. Enter `opkg update && opkg install python3-sqlite3`, restart the GUI and run the EPG import again (chapter 3).

The list is empty

Check the list address in the panel and choose **Refresh M3U list**. An Xstream list needs a correct server address with the port, login and password.

No EPG

Run **Import TV EPG** and wait for the completion message. When the import fails, change the *EPG source* or enter your own XMLTV address.

One channel has no schedule while the rest do

Assign a guide channel to it in **EPG matching** (chapter 10).

Black screen on exteplayer3 (5002)

Most often ServiceApp is not installed or does not match the system. Set *Player: GStreamer (4097)*. Open *Menu → Plugins* — if Enigma shows the „Some plugins are not available“ window with ServiceApp, install ServiceApp from your image's repository (chapter 3).

The picture freezes or a channel does not respond

The plugin tries to restore the picture by itself (chapter 7). When the problem returns on the same channels, try the other player (4097 ↔ 5002) and check the box's network connection — preferably wired.

The panel does not open

Check that the panel is switched on (*Settings → Web panel (PC)*), that the computer is in the same network and that the address is copied exactly. Restarting the box GUI also helps.

The plugin does not start after installation

Restart the GUI. Make sure the system uses Python 3 (Python 3.12 or 3.14, chapter 2) — this version does not work on images with Python 2.7.

15. Contact

The plugin is free. Comments, bug reports and ideas:

- e-mail: `hplink.soft@proton.me` — use the subject `HPL ink IPTV Enigma2`,
- Session: `noname554@`,
- other HPL ink programs: <https://hplspam1-jpg.github.io/HPL-ink-Software/index.html>.

HPL ink IPTV v2.01 · manual of 04.10.2026 · plugin tested on VU+ Duo 4K SE and VU+ Solo 4K with OpenBH 5.x (Python 3.12) and on OpenBH 6.0 (Python 3.14, with the python3-sqlite3 package).